When they are available, then, independence assertions can help in reducing the size of the domain representation and the complexity of the inference problem. Unfortunately, clean separation of entire sets of variables by independence is quite rare. Whenever a connection, however indirect, exists between two variables, independence will fail to hold. Moreover, even independent subsets can be quite large—for example, dentistry might involve dozens of diseases and hundreds of symptoms, all of which are interrelated. To handle such problems, we need more subtle methods than the straightforward concept of independence.

12.5 Bayes' Rule and Its Use

On page 390, we defined the **product rule** (Equation (12.4)). It can actually be written in two forms:

$$P(a \wedge b) = P(a \mid b)P(b) \qquad \text{and} \qquad P(a \wedge b) = P(b \mid a)P(a)\,.$$

Equating the two right-hand sides and dividing by $P(a)$, we get

$$P(b \mid a) = \frac{P(a \mid b)P(b)}{P(a)}\,. \tag{12.12}$$

This equation is known as **Bayes' rule** (also Bayes' law or Bayes' theorem). This simple equation underlies most modern AI systems for probabilistic inference. Bayes' rule

The more general case of Bayes' rule for multivalued variables can be written in the **P** notation as follows:

$$\mathbf{P}(Y \mid X) = \frac{\mathbf{P}(X \mid Y)\mathbf{P}(Y)}{\mathbf{P}(X)}\,.$$

As before, this is to be taken as representing a set of equations, each dealing with specific values of the variables. We will also have occasion to use a more general version conditionalized on some background evidence **e**:

$$\mathbf{P}(Y \mid X, \mathbf{e}) = \frac{\mathbf{P}(X \mid Y, \mathbf{e})\mathbf{P}(Y \mid \mathbf{e})}{\mathbf{P}(X \mid \mathbf{e})}\,. \tag{12.13}$$

12.5.1 Applying Bayes' rule: The simple case

On the surface, Bayes' rule does not seem very useful. It allows us to compute the single term $P(b \mid a)$ in terms of three terms: $P(a \mid b)$, $P(b)$, and $P(a)$. That seems like two steps backwards; but Bayes' rule is useful in practice because there are many cases where we do have good probability estimates for these three numbers and need to compute the fourth. Often, we perceive as evidence the *effect* of some unknown *cause* and we would like to determine that cause. In that case, Bayes' rule becomes

$$P(cause \mid effect) = \frac{P(effect \mid cause)P(cause)}{P(effect)}\,.$$

The conditional probability $P(effect \mid cause)$ quantifies the relationship in the **causal** direction, whereas $P(cause \mid effect)$ describes the **diagnostic** direction. In a task such as medical diagnosis, we often have conditional probabilities on causal relationships. The doctor knows $P(symptoms \mid disease)$) and want to derive a diagnosis, $P(disease \mid symptoms)$. Causal Diagnostic

For example, a doctor knows that the disease meningitis causes a patient to have a stiff neck, say, 70% of the time. The doctor also knows some unconditional facts: the prior

probability that any patient has meningitis is 1/50,000, and the prior probability that any patient has a stiff neck is 1%. Letting s be the proposition that the patient has a stiff neck and m be the proposition that the patient has meningitis, we have

$$\begin{aligned} P(s\,|\,m) &= 0.7 \\ P(m) &= 1/50000 \\ P(s) &= 0.01 \\ P(m\,|\,s) &= \frac{P(s\,|\,m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014\,. \end{aligned} \tag{12.14}$$

That is, we expect only 0.14% of patients with a stiff neck to have meningitis. Notice that even though a stiff neck is quite strongly indicated by meningitis (with probability 0.7), the probability of meningitis in patients with stiff necks remains small. This is because the prior probability of stiff necks (from any cause) is much higher than the prior for meningitis.

Section 12.3 illustrated a process by which one can avoid assessing the prior probability of the evidence (here, $P(s)$) by instead computing a posterior probability for each value of the query variable (here, m and $\neg m$) and then normalizing the results. The same process can be applied when using Bayes' rule. We have

$$\mathbf{P}(M\,|\,s) = \alpha\,\langle P(s\,|\,m)P(m), P(s\,|\,\neg m)P(\neg m)\rangle\,.$$

Thus, to use this approach we need to estimate $P(s\,|\,\neg m)$ instead of $P(s)$. There is no free lunch—sometimes this is easier, sometimes it is harder. The general form of Bayes' rule with normalization is

$$\mathbf{P}(Y\,|\,X) = \alpha\,\mathbf{P}(X\,|\,Y)\mathbf{P}(Y)\,, \tag{12.15}$$

where α is the normalization constant needed to make the entries in $\mathbf{P}(Y\,|\,X)$ sum to 1.

One obvious question to ask about Bayes' rule is why one might have available the conditional probability in one direction, but not the other. In the meningitis domain, perhaps the doctor knows that a stiff neck implies meningitis in 1 out of 5000 cases; that is, the doctor has quantitative information in the **diagnostic** direction from symptoms to causes. Such a doctor has no need to use Bayes' rule.

Unfortunately, *diagnostic knowledge is often more fragile than causal knowledge.* If there is a sudden epidemic of meningitis, the unconditional probability of meningitis, $P(m)$, will go up. The doctor who derived the diagnostic probability $P(m\,|\,s)$ directly from statistical observation of patients before the epidemic will have no idea how to update the value, but the doctor who computes $P(m\,|\,s)$ from the other three values will see that $P(m\,|\,s)$ should go up proportionately with $P(m)$. Most important, the causal information $P(s\,|\,m)$ is *unaffected* by the epidemic, because it simply reflects the way meningitis works. The use of this kind of direct causal or model-based knowledge provides the crucial robustness needed to make probabilistic systems feasible in the real world.

12.5.2 Using Bayes' rule: Combining evidence

We have seen that Bayes' rule can be useful for answering probabilistic queries conditioned on one piece of evidence—for example, the stiff neck. In particular, we have argued that probabilistic information is often available in the form $P(\mathit{effect}\,|\,\mathit{cause})$. What happens when we have two or more pieces of evidence? For example, what can a dentist conclude if her

nasty steel probe catches in the aching tooth of a patient? If we know the full joint distribution (Figure 12.3), we can read off the answer:

$$\mathbf{P}(Cavity \mid toothache \wedge catch) = \alpha \langle 0.108, 0.016 \rangle \approx \langle 0.871, 0.129 \rangle .$$

We know, however, that such an approach does not scale up to larger numbers of variables. We can try using Bayes' rule to reformulate the problem:

$$\begin{aligned} &\mathbf{P}(Cavity \mid toothache \wedge catch) \\ &\quad = \alpha \mathbf{P}(toothache \wedge catch \mid Cavity) \mathbf{P}(Cavity) . \end{aligned} \tag{12.16}$$

For this reformulation to work, we need to know the conditional probabilities of the conjunction $toothache \wedge catch$ for each value of *Cavity*. That might be feasible for just two evidence variables, but again it does not scale up. If there are n possible evidence variables (X rays, diet, oral hygiene, etc.), then there are $O(2^n)$ possible combinations of observed values for which we would need to know conditional probabilities. This is no better than using the full joint distribution.

To make progress, we need to find some additional assertions about the domain that will enable us to simplify the expressions. The notion of **independence** in Section 12.4 provides a clue, but needs refining. It would be nice if *Toothache* and *Catch* were independent, but they are not: if the probe catches in the tooth, then it is likely that the tooth has a cavity and that the cavity causes a toothache. These variables *are* independent, however, *given the presence or the absence of a cavity*. Each is directly caused by the cavity, but neither has a direct effect on the other: toothache depends on the state of the nerves in the tooth, whereas the probe's accuracy depends primarily on the dentist's skill, to which the toothache is irrelevant.[7] Mathematically, this property is written as

$$\mathbf{P}(toothache \wedge catch \mid Cavity) = \mathbf{P}(toothache \mid Cavity) \mathbf{P}(catch \mid Cavity) . \tag{12.17}$$

Conditional independence

This equation expresses the **conditional independence** of *toothache* and *catch* given *Cavity*. We can plug it into Equation (12.16) to obtain the probability of a cavity:

$$\begin{aligned} &\mathbf{P}(Cavity \mid toothache \wedge catch) \\ &\quad = \alpha \mathbf{P}(toothache \mid Cavity) \mathbf{P}(catch \mid Cavity) \mathbf{P}(Cavity) . \end{aligned} \tag{12.18}$$

Now the information requirements are the same as for inference, using each piece of evidence separately: the prior probability $\mathbf{P}(Cavity)$ for the query variable and the conditional probability of each effect, given its cause.

The general definition of **conditional independence** of two variables X and Y, given a third variable Z, is

$$\mathbf{P}(X, Y \mid Z) = \mathbf{P}(X \mid Z) \mathbf{P}(Y \mid Z) .$$

In the dentist domain, for example, it seems reasonable to assert conditional independence of the variables *Toothache* and *Catch*, given *Cavity*:

$$\mathbf{P}(Toothache, Catch \mid Cavity) = \mathbf{P}(Toothache \mid Cavity) \mathbf{P}(Catch \mid Cavity) . \tag{12.19}$$

Notice that this assertion is somewhat stronger than Equation (12.17), which asserts independence only for specific values of *Toothache* and *Catch*. As with absolute independence in Equation (12.11), the equivalent forms

$$\mathbf{P}(X \mid Y, Z) = \mathbf{P}(X \mid Z) \quad \text{and} \quad \mathbf{P}(Y \mid X, Z) = \mathbf{P}(Y \mid Z)$$

[7] We assume that the patient and dentist are distinct individuals.